

The elements

The building blocks of every single thing on Earth are pure chemical substances called elements. Put a few elements together by joining their atoms and you can get anything from a flea to a space rocket. Carbon-based elements are found in all living things, while water has just two elements – hydrogen and oxygen.

WHAT IS THE PERIODIC TABLE?

Elements are listed in a chart called the periodic table. Each entry shows the element's name, short chemical symbol, atomic number, and atomic mass.

ATOMIC NUMBER

An element's atomic number refers to the number of protons in the nucleus of an atom of the element. Titanium has 22.

22

47.867

ATOMIC MASS

This is the number of protons and neutrons inside an atom.

Ti

CHEMICAL SYMBOL

This scientific symbol is a short version of the element's name.

TITANIUM

NAME

This is the element's name. Titanium is a strong, light metal that is found in abundance in Earth's crust.

THE PERIODIC TABLE

Elements with a similar atomic structure sit together in the grid, which predicts how they will behave. Most of the 118 elements occur in rocks or in the atmosphere, but scientists have also built new ones by smashing smaller atoms together.

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1 1 1.0079 H Jupiter HYDROGEN	2 4 9.0122 Be Aquamarine BERYLLIUM	3 3 6.941 Li Watermelon tourmaline LITHIUM	11 3 22.990 Na Salt SODIUM	12 4 24.305 Mg Peridot MAGNESIUM	19 4 39.098 K Banana POTASSIUM	20 4 40.078 Ca Cheese CALCIUM	21 4 44.956 Sc SCANDIUM	22 4 47.867 Ti Benitoite gemstone TITANIUM	23 5 50.942 V Vanadinite VANADIUM	24 6 51.996 Cr Chrome tap CHROMIUM	25 7 54.938 Mn Spessartine MANGANESE	26 8 55.845 Fe Iron horseshoe IRON	27 9 58.933 Co Blue glass COBALT				
37 5 85.468 Rb Fireworks RUBIDIUM	38 5 87.62 Sr Strontium STRONTIUM	39 5 88.906 Y YTTRIUM	40 5 91.224 Zr Zircon stone ZIRCONIUM	41 5 92.906 Nb Rocket engine NIOBIUM	42 6 95.94 Mo Steel girder MOLYBDENUM	43 6 (96) Tc TECHNETIUM	44 6 101.07 Ru RUTHENIUM	45 6 102.91 Rh Rhodium-plated buckle RHODIUM	55 6 132.91 Cs Caesium atomic clock CAESIUM	56 6 137.33 Ba Crystals BARIUM	57-71 6 LANTHANIDE	72 6 178.49 Hf Nuclear reactor control rod HAFNIUM	73 6 180.95 Ta Capacitor TANTALUM	74 6 183.84 W Filament in light bulb TUNGSTEN	75 6 186.21 Re Jet turbine blades RHENIUM	76 6 190.23 Os Tip of fountain pen OSMIUM	77 6 192.22 Ir Surgical needle IRIDIUM
87 7 (223) Fr FRANCIUM	88 7 (226) Ra RADIUM	89-103 7 ACTINIDE	104 7 (261) Rf RUTHERFORDIUM	105 7 (262) Db DUBNIUM	106 7 (266) Sg SEABORGIUM	107 7 (264) Bh BOHRIUM	108 7 (277) Hs HASSIUM	109 7 (268) Mt MEITNERIUM	57 6 138.91 La Monazite LANTHANUM	58 6 140.12 Ce Cerium oxide CERIUM	59 6 140.91 Pr Permanent magnet PRASEODYMIUM	60 6 144.24 Nd Earphones NEODYMIUM	61 6 (145) Pm PROMETHIUM	62 6 150.36 Sm Samariskite SAMARIUM			
89 7 (227) Ac ACTINIUM	90 7 232.04 Th THORIUM	91 7 231.04 Pa PROTACTINIUM	92 7 238.03 U Nuclear power plant URANIUM	93 7 (237) Np NEPTUNIUM	94 7 (244) Pu Atomic bomb PLUTONIUM												

MAN-MADE ELEMENTS
Elements with a higher number than uranium (92) rarely occur naturally. These transuranic elements are created in particle accelerators or nuclear reactors. They include plutonium, which was used to make the atomic bomb.

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